

**Project Design Phase-II**  
**Technology Stack (Architecture & Stack)**

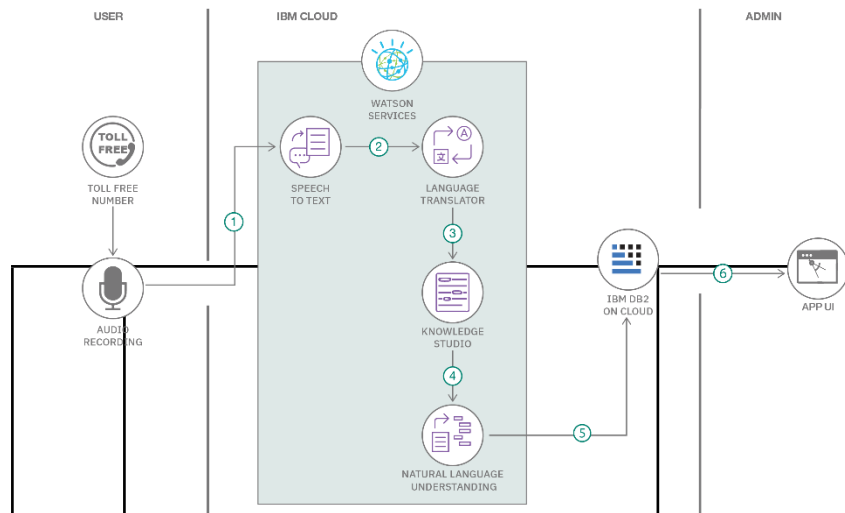
|               |                                                |
|---------------|------------------------------------------------|
| Date          | 30 september 2026                              |
| Team ID       | SWTID-2026-8393                                |
| Project Name  | Implement Client Script & UI Policy (Incident) |
| Maximum Marks | 4 Marks                                        |

**Technical Architecture:**

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example: Order processing during pandemics for offline mode**

**Reference:** <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



#### Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

| S.No | Component           | Description                                                               | Technology                                                     |
|------|---------------------|---------------------------------------------------------------------------|----------------------------------------------------------------|
| 1.   | User Interface      | How user interacts with application e.g. Web UI, Mobile App, Chatbot etc. | HTML, CSS, JavaScript / Angular Js / React Js etc.             |
| 2.   | Application Logic-1 | Logic for a process in the application                                    | Java / Python                                                  |
| 3.   | Application Logic-2 | Logic for a process in the application                                    | IBM Watson STT service                                         |
| 4.   | Application Logic-3 | Logic for a process in the application                                    | IBM Watson Assistant                                           |
| 5.   | Database            | Data Type, Configurations etc.                                            | MySQL, NoSQL, etc.                                             |
| 6.   | Cloud Database      | Database Service on Cloud                                                 | IBM DB2, IBM Cloudant etc.                                     |
| 7.   | File Storage        | File storage requirements                                                 | IBM Block Storage or Other Storage Service or Local Filesystem |
| 8.   | External API-1      | Purpose of External API used in the application                           | IBM Weather API, etc.                                          |

|     |                                 |                                                                                                               |                                        |
|-----|---------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 9.  | External API-2                  | Purpose of External API used in the application                                                               | Aadhar API, etc.                       |
| 10. | Machine Learning Model          | Purpose of Machine Learning Model                                                                             | Object Recognition Model, etc.         |
| 11. | Infrastructure (Server / Cloud) | Application Deployment on Local System / Cloud<br>Local Server Configuration:<br>Cloud Server Configuration : | Local, Cloud Foundry, Kubernetes, etc. |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description                                                                                                               | Technology      |
|------|-----------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| 5.   | Performance     | Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc. | Technology used |

**References:**

| S.No | Characteristics          | Description                                                                                    | Technology                                          |
|------|--------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1.   | Open-Source Frameworks   | List the open-source frameworks used                                                           | Technology of Opensource framework                  |
| 2.   | Security Implementations | List all the security / access controls implemented, use of firewalls etc.                     | e.g. SHA-256, Encryptions, IAM Controls, OWASP etc. |
| 3.   | Scalable Architecture    | Justify the scalability of architecture (3 – tier, Micro-services)                             | Technology used                                     |
| 4.   | Availability             | Justify the availability of application (e.g. use of load balancers, distributed servers etc.) | Technology used                                     |

<https://c4model.com/> <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/> <https://www.ibm.com/cloud/architecture> <https://aws.amazon.com/architecture>  
<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>